

Class #5 - Linear Combination

1. Def: Let $u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$, and $v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$. Then u and v are _____ in _____.

The *algebra* of vectors are defined as follows:

(a)

$$u + v = \text{_____} \in \mathbf{R}^n$$

(Addition)

(b)

$$cu = \text{_____} \in \mathbf{R}^n$$

(Scalar Multiplication)

2. Properties of Addition and Scalar Multiplication:

(a) $u + v = \text{_____}$

(b) $(u + v) + w = \text{_____}$

(c) $u + 0 = \text{_____} = u$ where 0 denotes the _____ in $\mathbf{R}^n$.

(d) $u + (-u) = -u + u = \text{_____}$ where $-u$ denotes the vector _____.

(e) $c(u + v) = \text{_____}$ where c is a _____.

(f) $(c + d)u = \text{_____}$ where c, d are _____.

(g) $c(du) = \text{_____}$

(h) $1u = \text{_____}$

3. Example: Let $u = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, and $v = \begin{bmatrix} -3 \\ -1 \end{bmatrix}$.

$u + v =$ _____

$u - 2v =$ _____

$0u =$ _____ where 0 denotes _____

4. Fill in the blank: $u = v$ if and only if _____.

5. Def: Given a finite number of vectors $u_1, u_2, \dots, u_p$ in $\mathbf{R}^n$, we say that y is a *linear combination* of $u_1, u_2, \dots, u_p$ with weights $x_1, \dots, x_p$ if the _____ equation

$$y = x_1 u_1 + x_2 u_2 + \dots + x_p u_p$$

is true for *some* _____ $x_1, \dots, x_p$.

6. Examples:

(a) The vector $y = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$ _____ a linear combinations of u and v in #3.

(b) Let $u_1, u_2, \dots, u_p$ be in $\mathbf{R}^n$. Is u_1 a linear combination of $u_1, \dots, u_p$?

What are the weights?

What about u_2 ?

Is 0 a linear combination of $u_1, \dots, u_p$? (what's 0?)

Is cu_1 a linear combination of $u_1, \dots, u_p$ for any scalar c ?

What are the weights?

(c) Fill in the blank: _____ is a linear combination of *any* given vectors in $\mathbf{R}^n$.

7. Example: Write the following _____ equation as a linear system:

$$\begin{bmatrix} 3 \\ -7 \\ -3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}.$$

8. Observations:

- (a) Given a linear system, we can write a corresponding _____ equation.
- (b) A linear combination of vectors in $\mathbf{R}^n$ is in _____.
- (c) A vector equation solving for the weights is the same thing for solving a linear system for the variables _____.

9. Let $y = \begin{bmatrix} 3 \\ -7 \\ -3 \end{bmatrix}$, $u_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$, $u_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, and $u_3 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$.

Is y a linear combination of the vectors u_1, u_2, u_3 ? If so, what are the weights?

10. Fill in the blank: y is a linear combination of the vectors u_1, u_2, u_3 if and only if the corresponding linear system is _____.